Date	1/10/2026	Maximum Marks	4 Marks
Team ID	SWTID-2026-4326		
Project Name	Streamlining IT Procurement automating laptop orders with flow designer		

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Functional Requirements:

Following are the functional requirements of the proposed automated IT procurement solution, combining **User Perspectives** and **Designer Execution Views**.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Catalog Browsing & Laptop Selection	• [User View]: Role-based hardware catalog browsing (Developer, Manager, General Staff). • [Designer View]: Dynamic card grid displaying side-by-side spec comparison (RAM, CPU, Weight). • Filter laptops by department eligibility and current stock availability.
FR-2	Automated Request Submission	• [User View]: Single-click ordering pre-populated with employee profile data. • [Designer View]: Integration with Workday API to auto-fill Employee ID, Cost Center, and Manager. • Validation checking to prevent duplicate active laptop requests.
FR-3	Multi-Channel Approval Routing	• [User View]: Automated manager approval via Email or MS Teams notification. • [Designer View]: Flow Designer rule engine triggering conditional logic (Auto-approve standard models under budget threshold). • Automated 24-hour escalation timer if manager approval is delayed.
FR-4	Vendor Order Provisioning	• [User View]: Instant confirmation email with estimated delivery date upon sign-off. • [Designer View]: Automated REST API payload transmission to hardware vendor ERP. • Automatic Purchase Order (PO) creation in internal finance systems.
FR-5	Real-Time Delivery & Asset Tracking	• [User View]: Live visual progress bar showing stage-by-stage status (Submitted → Approved → Shipped → Delivered). • [Designer View]: Integration with courier webhook API for live package tracking link. • Asset management database sync upon user delivery sign-off.
FR-6	Admin Control & Analytics Dashboard	• [User View]: Executive summary showing total procurement cycle time reduction. • [Designer View]: ServiceNow dashboard reporting SLA compliance, bottleneck logs, and active stock counts.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution detailing performance targets, security standards, and system availability.

NFR No.	Non-Functional Requirement	Description & Engineering Specifications
NFR-1	**Usability**	• The self-service catalog UI must require **no more than 3 clicks** from login to final order submission. • Responsive design supporting desktop web browsers and ServiceNow mobile app views. • Clear error messaging and real-time form field validation indicators.
NFR-2	**Security & Compliance**	• Single Sign-On (SSO) authentication using OAuth2 / SAML 2.0 enterprise credentials. • Role-Based Access Control (RBAC) ensuring users only access eligible hardware tiers. • End-to-end data encryption in transit (TLS 1.3) and at rest (AES-256) for employee records.
NFR-3	**Reliability**	• Automated retry mechanisms in Flow Designer for failed vendor REST API calls. • Zero data loss for pending order workflows during scheduled ServiceNow platform maintenance. • System audit logging recording every approval action with timestamp and IP address.
NFR-4	**Performance**	• Service Catalog page load time must be **under 1.5 seconds** under peak load. • Flow Designer trigger execution time under 2 seconds upon order submission. • Real-time webhook notifications for approval status updates within 5 seconds.
NFR-5	**Availability**	• System availability SLA of **99.9% uptime** (24/7/365 access). • Scheduled maintenance restricted to non-business weekend hours with 48-hour prior notification.
NFR-6	**Scalability**	• Capable of handling up to **10,000 concurrent laptop order requests** during peak company onboarding waves. • Modular Flow Designer structure allowing seamless addition of new hardware types (monitors, mobiles) without code refactoring.